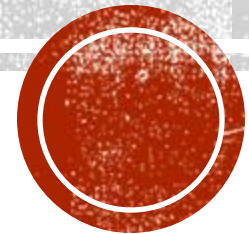


WEEK 4

Linear Classification

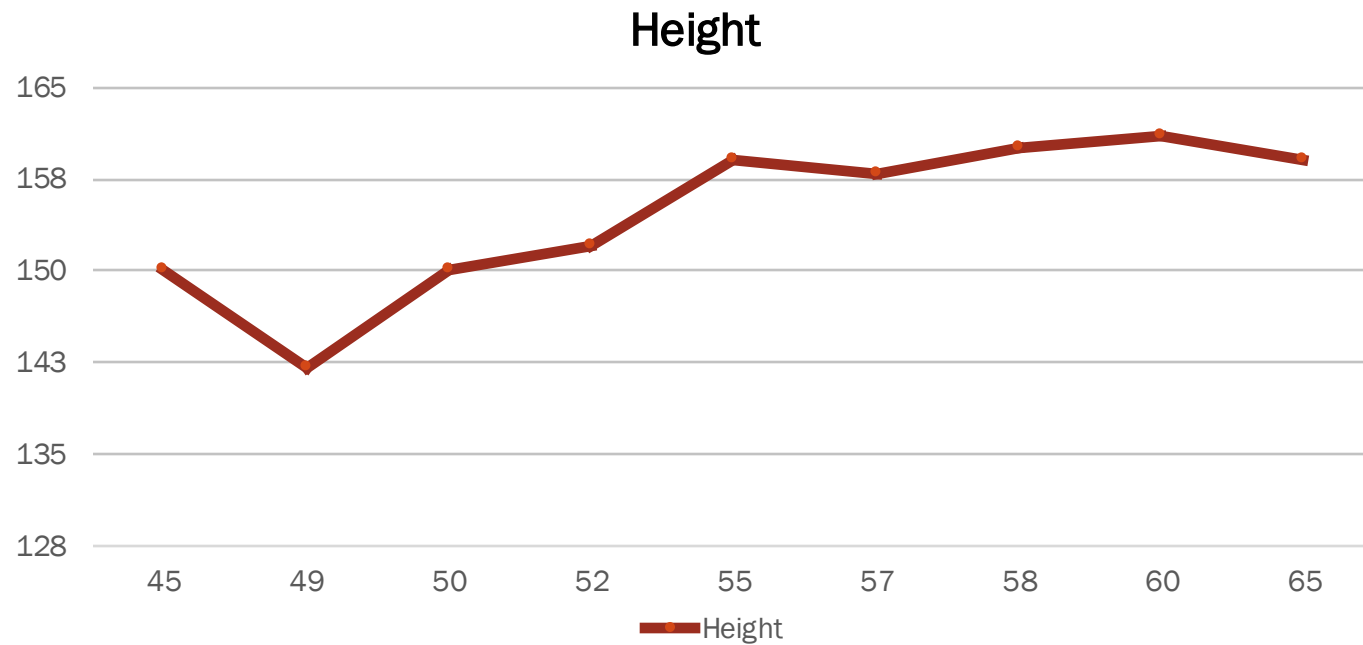


LINEAR MODELS

- Linear Regression
 - Prediction
- Linear Classification
 - Linear Discrimination
 - Logistic Regression



SIMPLE REGRESSION



BEST FIT METHOD(1)

- Minimising SSE (regression way)

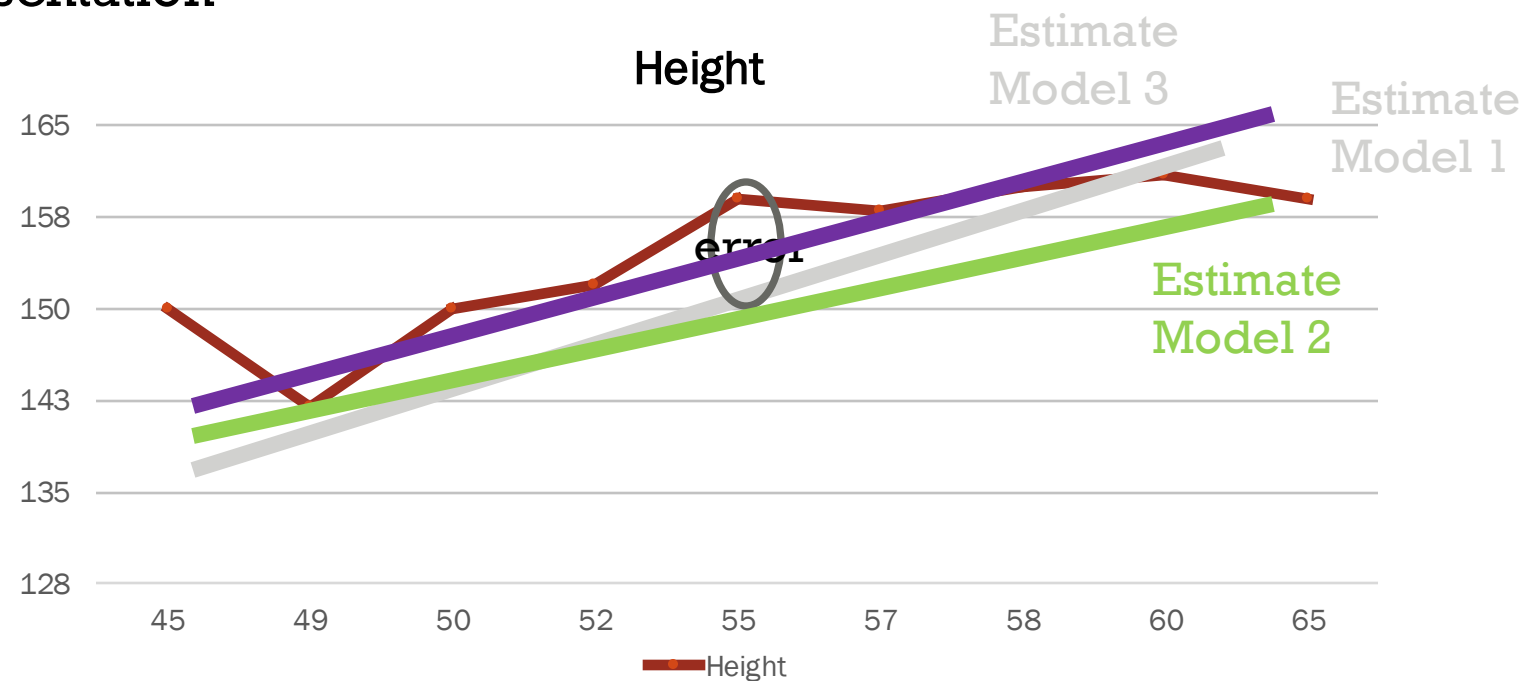
$$E = \frac{1}{2} (y(x; w) - t)^2$$

- $y(x; w)$ is your current linear function
- t is your ideal target
- **Minimising** the error ∂E
 - $\partial E / \partial w_j = 0$
 - $((w\mathbf{x} + w_0 - \mathbf{t})\mathbf{x}) = 0$
 - Using normal linear equations or other advanced techniques



SIMPLE LINEAR REGRESSION

- One input
- Fit the line
- Linear representation



REGRESSION IDEAS

$$y' = b + w_1 x_1$$

Prediction Bias Weight Feature value

Calculated from training

```
graph TD; P[Prediction] --> y_prime; B[Bias] --> b; W[Weight] --> w1; FV[Feature value] --> x1; B --- BW; W --- BW; BW --- CFT[Calculated from training];
```



FITTING THE MODEL

- The best fit of the model
- The most common way is to **‘reduce the errors’**
- **Simple linear model**

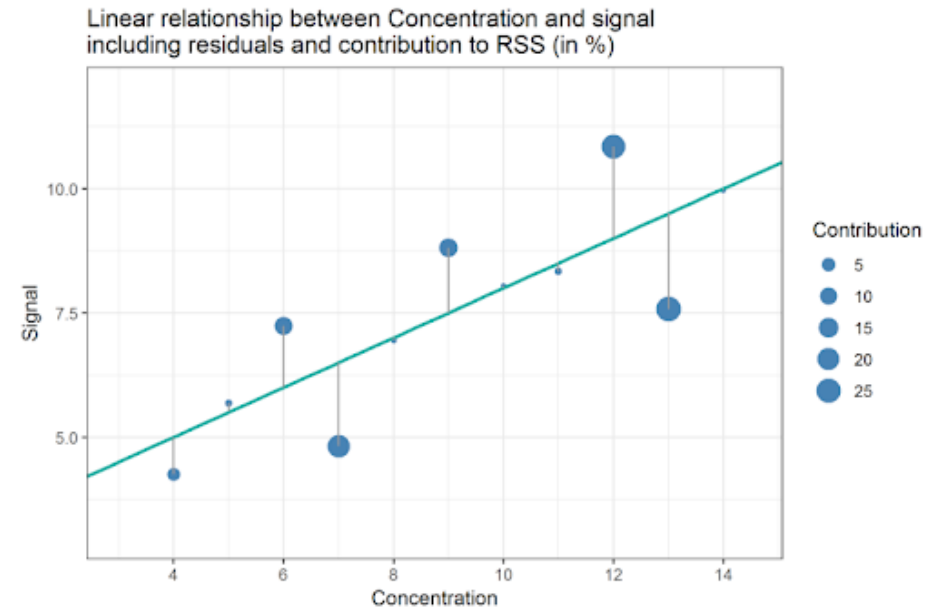
$$y' = b + w_1 x_1$$

w_1 $\odot$ Slope

b $\odot$ Intercept

- **Residual sum of square errors(RSS)**

$$\sum_{i=1}^n (y' - y)^2$$



ESTIMATE W-H

W	H	$H=W+99$	$H=0.5W+130$
55	159		
57	158		
58	160		
60	161		
65	159		

Which one fit better?

$H=W+99$ **and** $H=.5W+130$



MORE LOSS FUNCTION/EVALUATION

- Sum Absolute Error

$$SAE = \sum_{i=0}^n |Y - f(x)|$$

- Sum Squared Error

$$SSE = \sum_{i=0}^n (f(x) - y)^2$$

- Root mean square error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=0}^n (f(x) - y)^2}$$

**“Remember that all model are wrong; the practical question is how wrong do they have to be to not be useful”
George Box 1987**



EXAMPLE 2 (MULTI DIMENSIONAL)

Bedroom	Bathroom	Price(10K)
3	1	22.2
3	2	53.8
2	1	18

$$\text{Price} = w_1 \text{Bed} + w_2 \text{Bath} + c$$



LINEAR CLASSIFICATION/DISCRIMINATION

- In previous lectures, Bayes theorem are used for deciding class
- Linear regression can be implemented for classification/discrimination
- In practiced, a discriminant function is usually be defined



LINEAR DISCRIMINATION FUNCTIONS

- In Bayes, density estimation is made by using probability

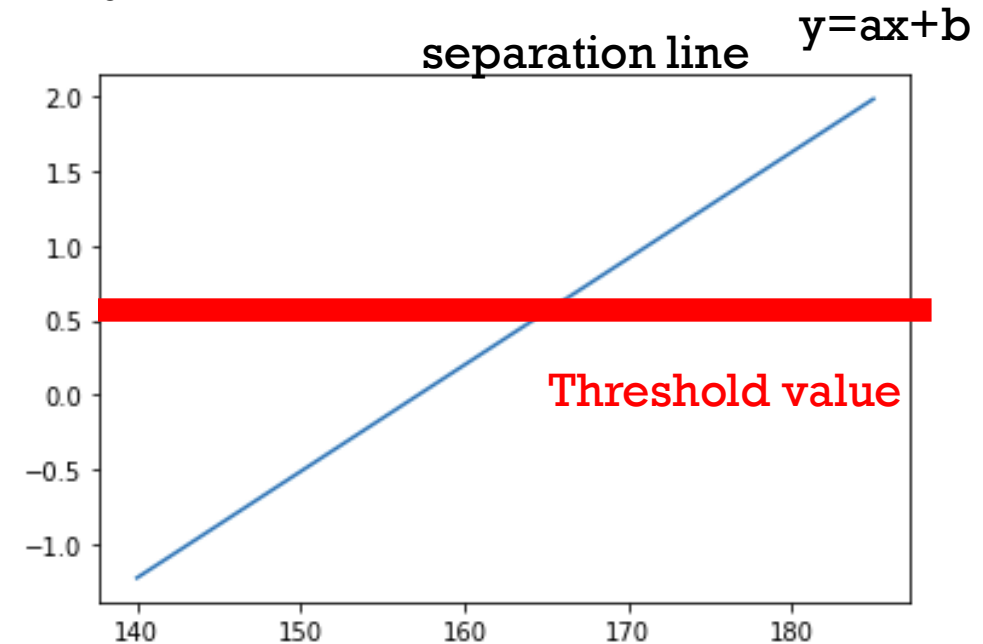
$$p(C_k | \mathbf{x}) = \frac{p(\mathbf{x} | C_k) p(C_k)}{p(\mathbf{x})}$$

- The linear discriminant function is defined as

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$

Two classes

- $\mathbf{w}$ is a weight vector
- w_0 is a bias, the negative bias is sometimes called *threshold*
- The input is classified to C_1 if $y(\mathbf{x}) \geq 0$ and class C_2 o.w.
- The boundary is defined at $y(\mathbf{x}) = 0$



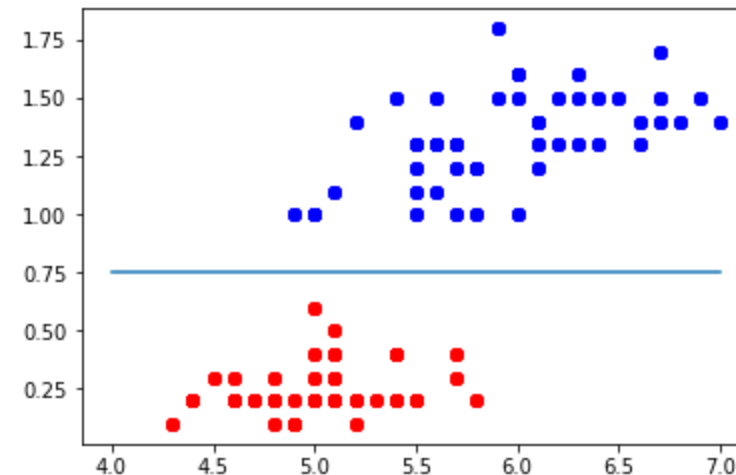
TWO CLASS DISCRIMINATION FUNCTION

- Two main equations

$$\mathbf{w}^T \mathbf{x} + w_0 \geq \theta \text{ For all } x \in C1$$

$$\mathbf{w}^T \mathbf{x} + w_0 < \theta \text{ For all } x \in C2$$

- Linear discriminant divides the feature space by a hyperplane
- The orientation is defined by w
- The location is defined by w_0



EXAMPLE

Women	Men
150	160
154	170
160	165

Find the best line to separate between them

Height	Class (data type)
150	1
154	1
160	1
160	0
170	0
165	0



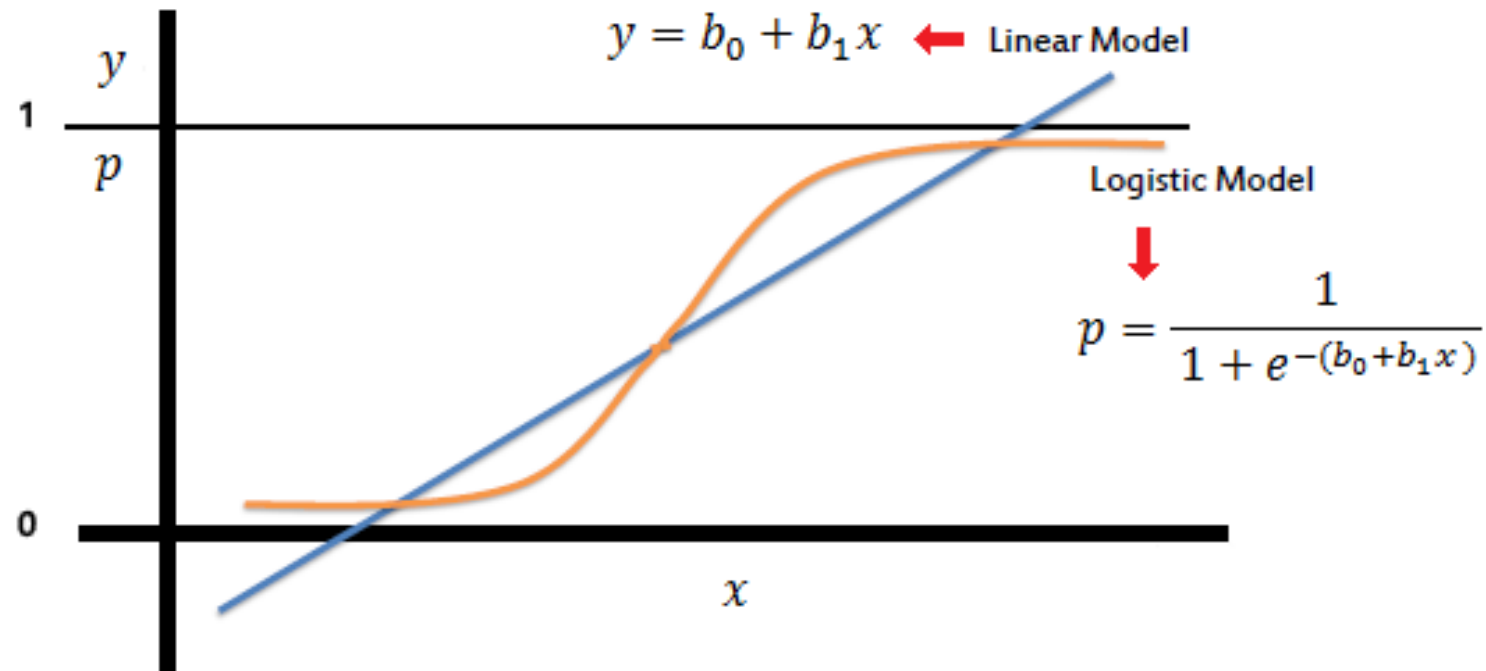
EXAMPLE: FINDING THE ERROR

Height	Y(output)	Class (t)
150	0.5	1
154	0.8	1
160	0.9	1
160	0.9	0
170	0.2	0
165	0.1	0

$$\text{Error} = 0.5(0.5^2 + 0.2^2 + 0.1^2 + 0.9^2 + 0.2^2 + 0.1^2) = 0.58$$



LINEAR $\rightarrow$ LOGISTIC MODEL



CROSS ENTROPY: TWO CLASS PROBLEMS

- Sum-of-squares is more suitable for **regression** because of the target
- More appropriate error estimation of **classification**

For network with a single output, representing 2 classes

- Probability to be class 1 $y=P(C_1 | \mathbf{x})$
- Probability to be class 2 $P(C_2 | \mathbf{x})=1-y$
- t is a target coding, $t=1$ means its belong to class 1 and 0 otherwise

$$-\sum t \log(p) + (1 - t) \log(1 - p)$$



BEST FIT METHOD(2)

- Cross entropy (binary)
 $-\sum t \log(p) + (1 - t) \log(1 - p)$

Height	p(c=1)	Class (data type)
150	0.5	1
154	0.8	1
160	0.9	1
160	0.9	0
170	0.2	0
165	0.1	0

$$-(\log(0.5) + \log(0.8) + \log(0.9) + \log(0.1) + \log(0.8) + \log(0.2)) = 0.04$$



MULTIDIMENSIONAL PROBLEM

Women		Men	
Weight	Height	Weight	Height
50	150	60	160
54	154	67	170
50	160	60	165

$$\text{Gender} = w_1 \text{Weight} + w_2 \text{Height} + c$$



DISCUSSION

- How to train?
- What can be a problem?



SK-LEARN LIBRARY

- Create
 - **from** sklearn.linear_model **import** LogisticRegression
 - **model = LogisticRegression()**
- Train
 - **model=model.fit(X, y)**
- Result
 - **y_pred=model.predict(X_test)**
 - **y_probs = model.predict_proba(X_test)**
 - **y_probs = model.log_proba(X_test)**



USING THE LIBRARY

Create a model

```
from sklearn import linear_model  
model = linear_model.LogisticRegression()
```

Train the model

```
model.fit(X_train, y_train)
```

Get the result

- `y_pred=model.predict(X_test)`
- `y_probs = model.predict_proba(X_test)`
- `y_probs = model.log_proba(X_test)`

Get the score

```
model.score(X_test, y_test)
```



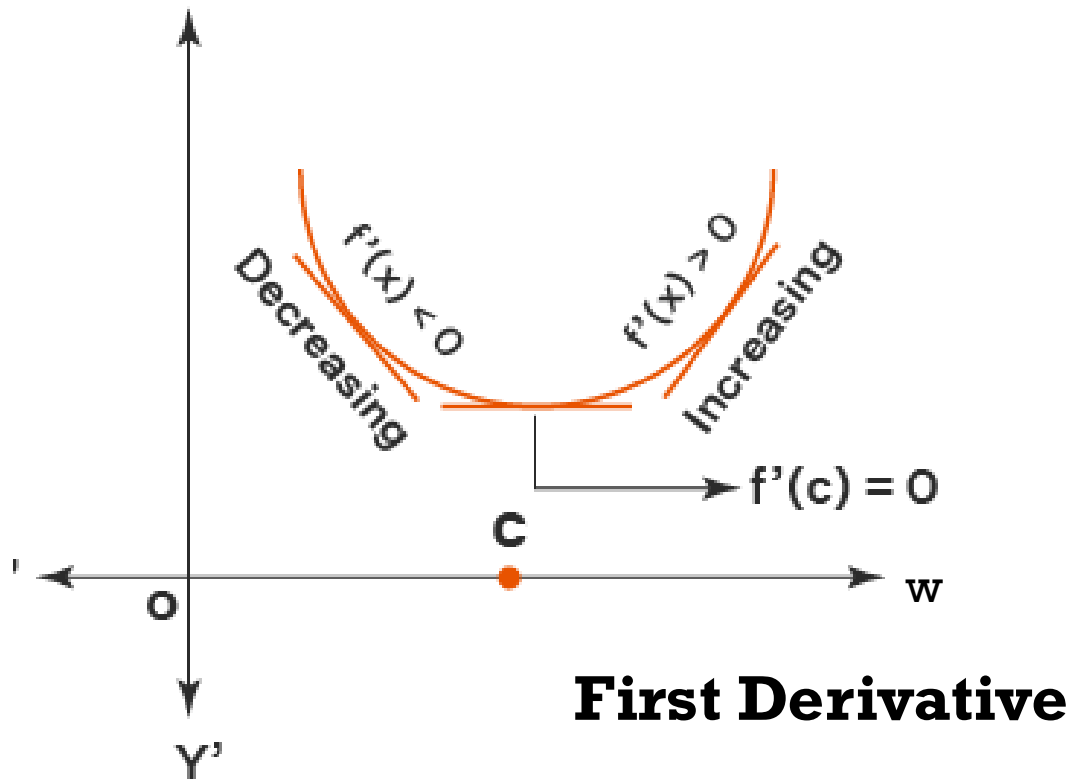
ATTRIBUTES

- **classes_** A list of class labels known to the classifier.
- **coef_** Coefficient of the features in the decision function.
- **intercept_** Intercept (a.k.a. bias) added to the decision function.



OPTIMIZATION

Error/ Loss function



Local Minima

Second Derivative Test

Local Max

$$f'(c) = 0 \text{ and } f''(c) < 0$$



Local Min

$$f'(c) = 0 \text{ and } f''(c) > 0$$

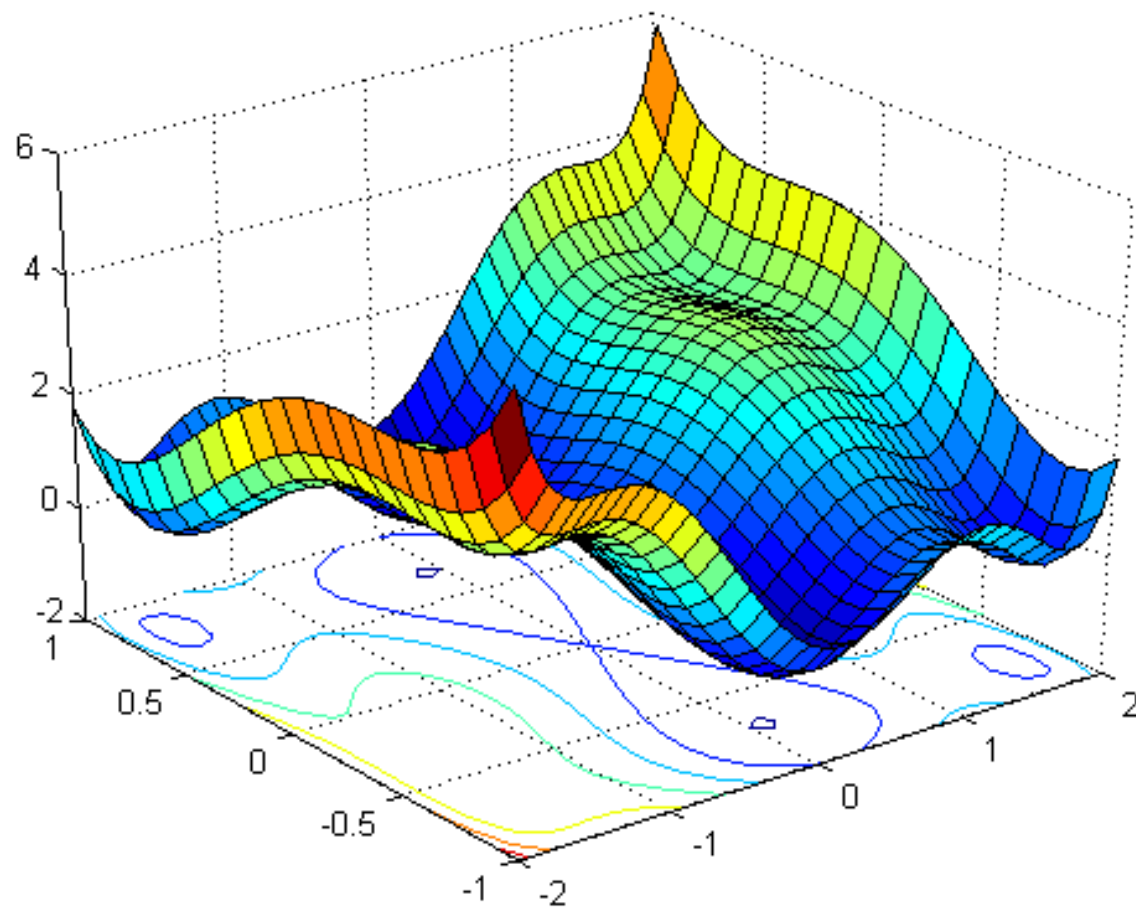


FINDING THE BEST W ?

- If the derivative is positive???
- If the derivative is negative???
- If the derivative is 0?



REAL WORLD DATA?



EVALUATION METHODS (REVISIT)

Confusion Matrix is usually used to understand the output values

		Predicted	
		+	-
Actual	+	TP	FN
	-	FP	TN



EXAMPLE

- What are
 - True Positive (TP)
 - False Positive (FP)
 - True Negative (TN)
 - False Negative (FN)

Negative(TN)

Negative(FN)

		Predicted	
		F(+)	M(-)
Actual	F(+)	5	5
	M(-)	4	6



SIMPLE EVALUATION MATRICES

- Overall accuracy

- $$\frac{TP+TN}{TP+FN+FP+TN} = \frac{5+6}{5+5+4+6} = 0.55 = 55\%$$

- Precision (prediction capabilities)

- Precision(F) = $5/9 = 0.55$
- Precision(M) = $6/11 = 0.545$

- Recall

- Recall(F) = $5/10 = 0.5$
- Recall(M) = $6/10 = 0.6$

		Predicted		RECALL
		F(+)	M(-)	
Actual	F(+)	5	5	
	M(-)	4	6	
		PRECISION		



MICRO-AVERAGE

- Micro-average

- Recalculate manually
- Micro-average precision

- $$\frac{TP_F + TPM}{TP_F + TPM + FP_F + FP_M}$$

- $$\frac{5+6}{5+6+4+6} = 0.52$$

- Micro-average Recall (Same value)

- $$\frac{TP_F + TPM}{TP_F + TPM + FP_F + FP_M}$$

- $$\frac{5+6}{5+6+4+6} = 0.52$$

Predicted		
F(+)	M(-)	
Actual	F(+)	5
	M(-)	6

PRECISION

RECALL



MACRO-AVERAGE

- Average of all classes
- Macro-average precision

- $\frac{\text{Precision}_M + \text{Precision}_F}{2}$

- $\frac{0.55 + 0.545}{2} = 0.5475$

- Macro-average Recall

- $\frac{\text{Recall}_M + \text{Recall}_F}{2}$

- $\frac{0.5 + 0.56}{2} = 0.55$

		Predicted		R E C A L L
		F(+)	M(-)	
Actual	F(+)	5	5	
	M(-)	4	6	
		PRECISION		



TPR/TNR

True positive rate(TPR/ sensitivity)

- The rate of true positive from all actual positive values
- Example $TPR = 5 / 10 = 0.5$ or 50% $= \frac{TP}{TP+FN}$

Predicted		
F(+)	M(-)	
Actual	F(+)	5
	M(-)	4
		TPR
		TNR

True negative rate(TNR / Specificity)

- The rate of true negative from all actual negative values
- Example $TNR = 6 / 10 = 0.6$ or 60% $= \frac{TN}{TN+FP}$



FPR/FNR

False negative rate(FNR)

- The rate of false negative from all actual positive values
- $FNR = \frac{FN}{FN+TP} = 1-TPR$
- Example $FNR = 1-TNR = 1-0.5 = 0.5$ or 50%

False positive rate(FPR/ Fall-out)

- The rate of false positive from all actual negative values
- $FPR = \frac{FP}{FP+TN} = 1-TNR$
- Example $FPR = 1-TNR = 1-0.6 = 0.4$ or 40%

Predicted		
F(+)	M(-)	
Actual	F(+)	5
	M(-)	4
		5
		6
		FNR=1-TPR
		FPR=1-TNR



PRECISION / RECALL

False negative rate(FNR)

- The rate of false negative from all actual positive values
- $FNR = \frac{FN}{FN+TP} = 1-TPR$
- Example $FNR = 1-TNR = 1-0.5 = 0.5$ or 50%

False positive rate(FPR/ Fall-out)

- The rate of false positive from all actual negative values
- $FPR = \frac{FP}{FP+TN} = 1-TNR$
- Example $FPR = 1-TNR = 1-0.6 = 0.4$ or 40%

Predicted		
F(+)	M(-)	
Actual	F(+)	5
	M(-)	4
		5
		6
		FNR=1-TPR
		FPR=1-TNR



F-MEASURE

- F_1 is a harmonic mean between precision and recall
- $F_1 = 2 \frac{\text{recall} \cdot \text{precision}}{\text{recall} + \text{precision}}$
- Best used with information retrieval and Natural Language Processing

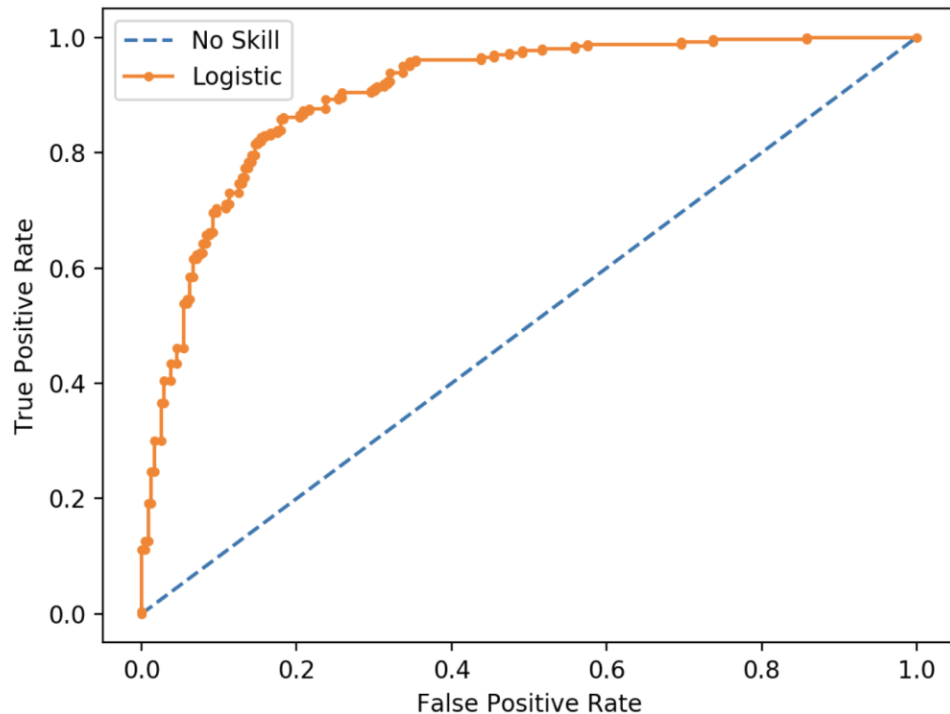


SUMMARY

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Ref: <https://www.ryangineer.com/>

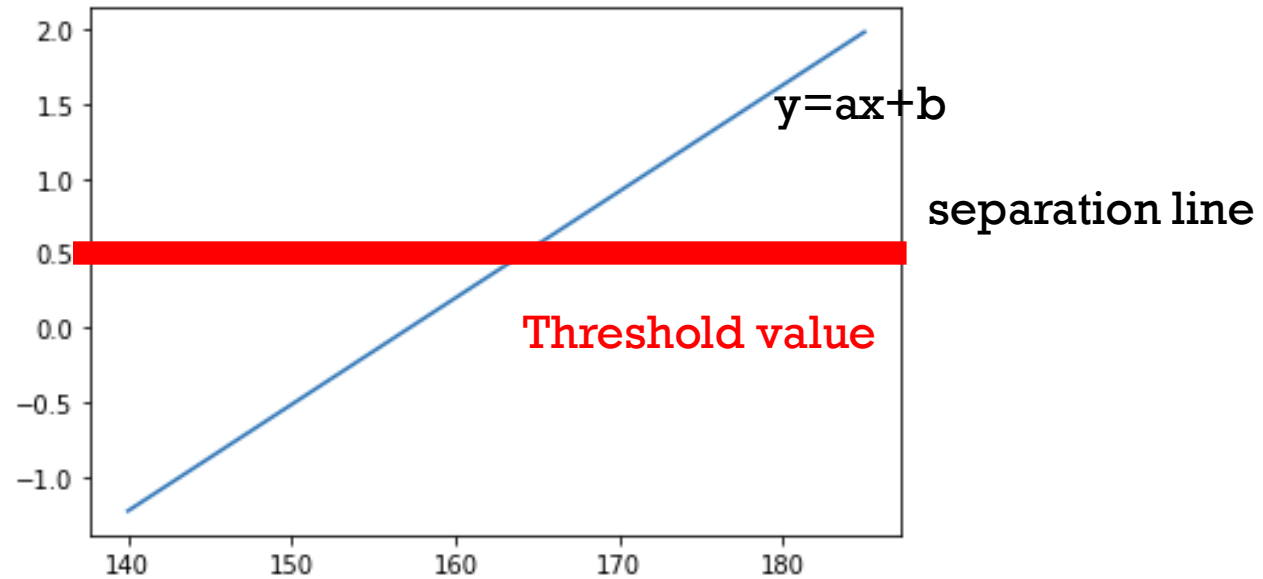
ROC CURVE



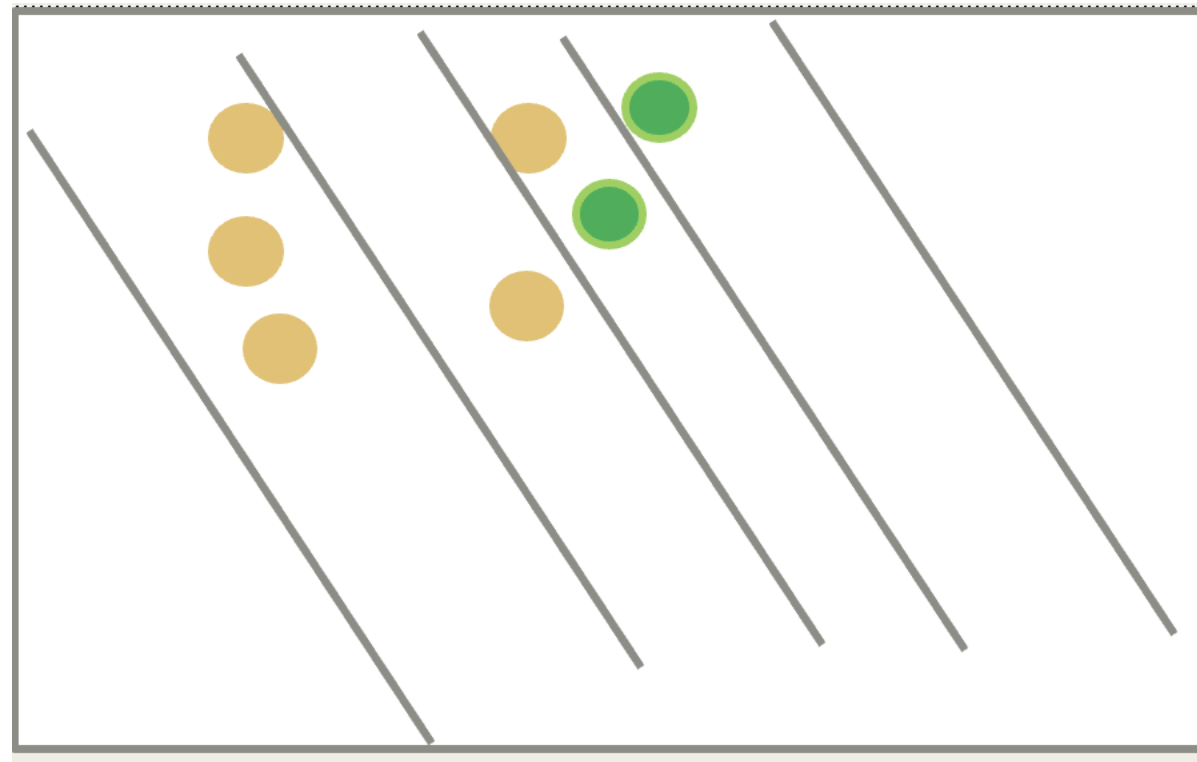
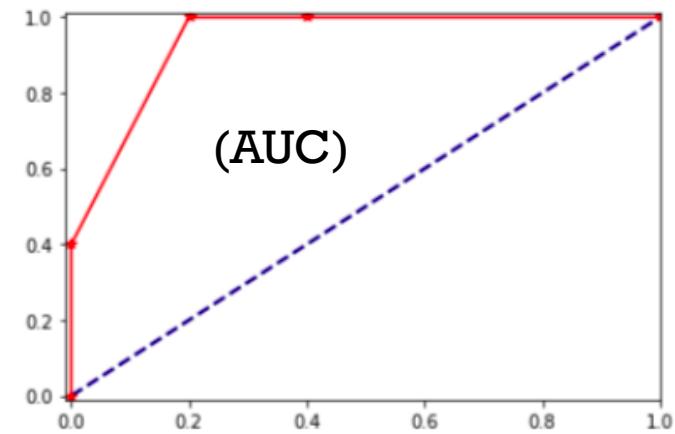
- Receiver operating characteristic curve
- A plot between TPR(y-axis) and FPR(x-axis)
- Area under curve (AUC) can be used to select better models



THRESHOLD VALUE



ROC CURVE

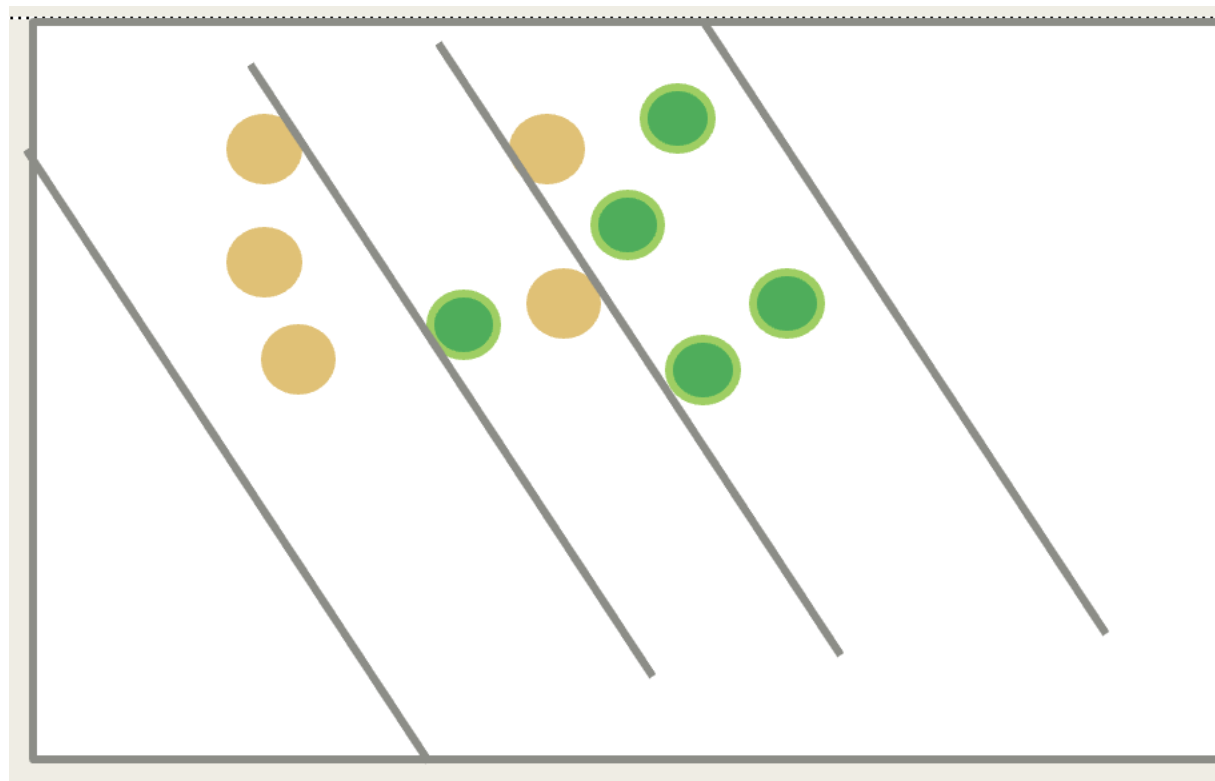


Green = +
Yellow = -

TPR	1	1	1	0.4	0
FPR	1	0.4	0.2	0	0

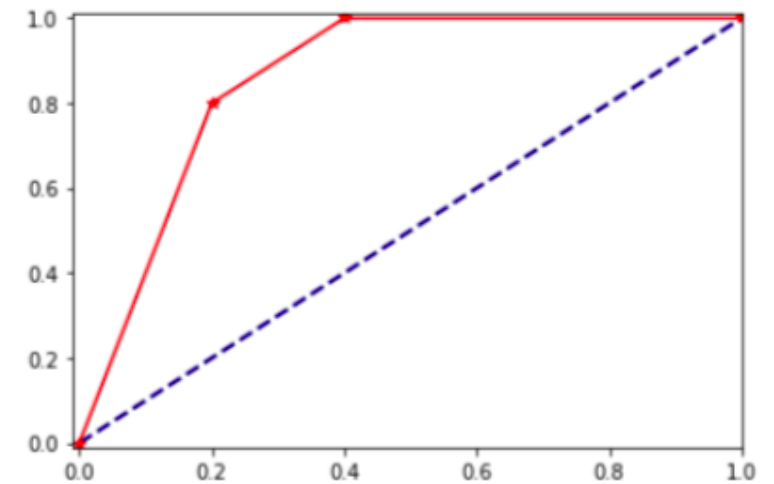


ROC CURVE



TPR
FPR

1 1
0.4 0.2
0 0

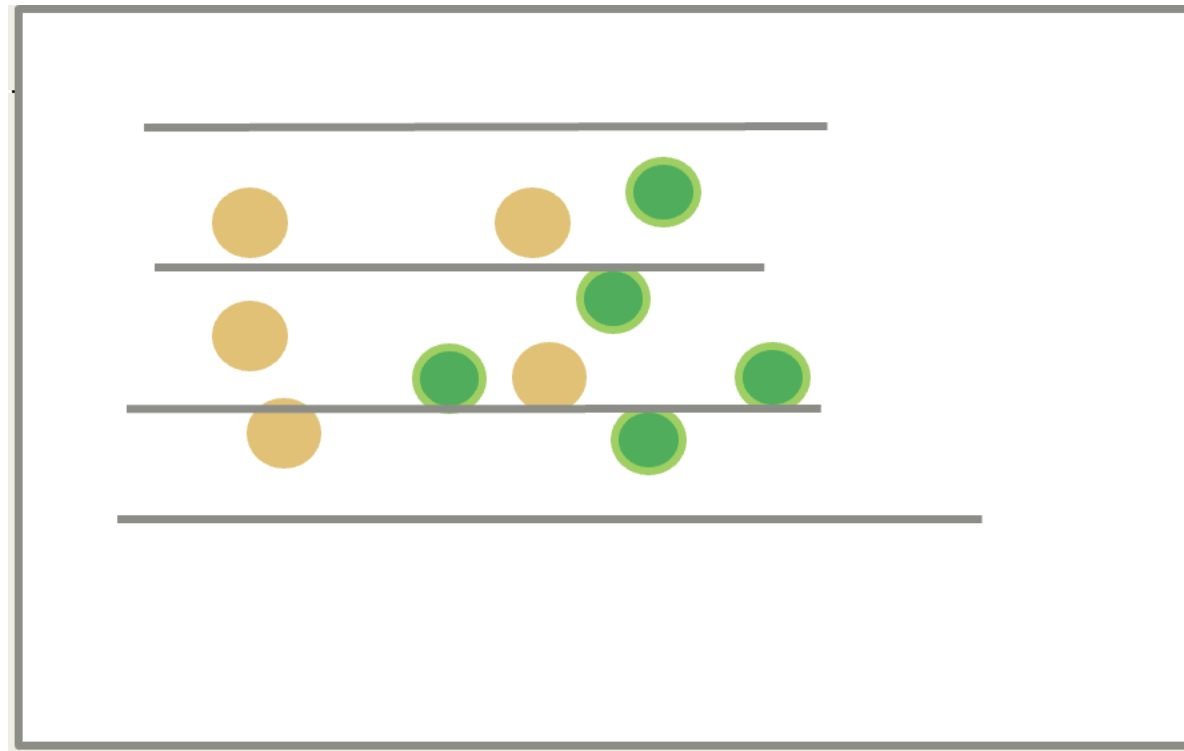
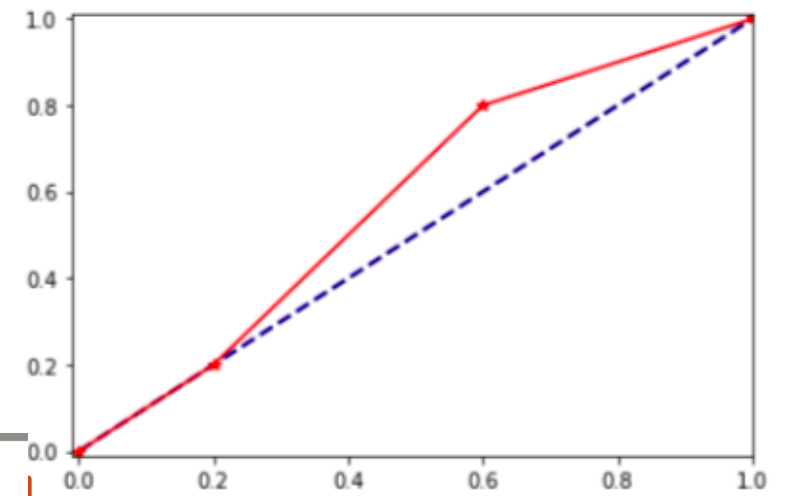


Green = +
Yellow = -



ROC CURVE

under the line make positive value



Green = +
Yellow = -

TPR	1	0.8	0.2	0
FPR	1	0.6	0.2	0



SK-LEARN LIBRARY (2)

```
from sklearn.metrics import precision_recall_curve
```

```
from sklearn.metrics import f1_score
```

```
from sklearn.metrics import auc
```

- `y_probs, lr_recall, _ = precision_recall_curve(y_test, y_probs )`
- `lr_f1= f1_score(y_test, y_pred)`
- `lr_auc= auc(lr_recall, lr_precision)`

